FoodFlow: Scalable Microservices Architecture

FoodFlow: Design & Development of a Scalable Microservices Architecture

Course: Scalable Services (SEZG583) | MTech
Assignment: Assignment 1 (13 Marks)
Team: Group 26
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Repository: https://github.com/kush6542/Group26_Scalable_Services

1. Problem and Scope

1.1 Problem Overview

Modern food delivery platforms operate under extreme transactional concurrency, transient network unreliability, and cross-domain business workflows. A customer order is not a simple database update; it represents a distributed multi-service journey involving inventory locking, monetary transactions, logistics dispatching, and asynchronous customer notifications.

FoodFlow is a distributed microservices platform designed from first principles to solve these challenges. It eliminates single points of failure, avoids distributed deadlocks, guarantees data consistency without cross-database transactions, and handles concurrent bursts without resource over-allocation.

1.2 System Assumptions

1. Each microservice operates within its own bounded context and executes against its own isolated relational database (**Database-per-Service**).
2. Inter-service communication occurs synchronously over HTTP/REST using JSON payloads and standardized error representations.
3. Network calls between services are prone to latency and partial failure; all distributed transactions must provide explicit compensating rollbacks.
4. Downstream third-party providers (payment gateways, SMS providers) are mocked with controllable latency and failure injection for verification.

1.3 Core Business Rules (Traceability Matrix)

Rule ID	Service	Rule Statement	Enforcement Mechanism	Test Name
BR1	Restaurant Service	An order item cannot be reserved if requested quantity exceeds available stock. Simultaneous requests must never over-allocate stock.	Atomic conditional SQL update (`UPDATE menu_items SET available_quantity = available_quantity - :qty WHERE available_quantity >= :qty`)	`test_br1_stock_sufficiency_`
BR2	Payment Service	Repeating an unsafe request with an identical `Idempotency-Key` and matching payload must return the original cached response with no duplicate charge.	In-memory/DB idempotency table storing SHA-256 payload hash and cached JSON response	`test_br2_payment_idempoten`
BR3	Payment & Order Services	Submitting an existing `Idempotency-Key` with an altered payload must be rejected with HTTP 409 Conflict.	SHA-256 fingerprint comparison at request entry	`test_br3_payment_idempoten`
BR4	Payment Service	Payment charges must have an amount > 0 and valid card details; test card ending in `9999` triggers simulated payment decline (HTTP 402).	Card validation logic & failure simulator	`test_payment_decline_simul`
BR5	Delivery Service	A delivery task can only be assigned to a rider whose status is `AVAILABLE` . If all riders are busy, returns HTTP 409 Conflict.	Rider status filtering & atomic status lock to `BUSY`	`test_br5_rider_assignment_`
BR6	Order Service	A multi-service checkout transaction	Orchestrated Saga engine with	`test_order_checkout_happy_`

Rule ID	Service	Rule Statement	Enforcement Mechanism	Test Name
		must transition through explicit states (`PENDING` → `STOCK_RESERVED` → `PAID` → `DELIVERY_ASSIGNED` → `CONFIRMED`).	persistent state transition logging	
BR7	Order Service	If a checkout step fails downstream, all preceding successful operations must be rolled back via compensating calls.	SAGA Compensator issuing reverse actions (refund, stock release, task cancel)	`test_order_checkout_payment`

1.4 Out of Scope

- User authentication and JWT authorization (simulated via client headers).
- Front-end UI (all interactions driven via REST APIs, Swagger UI, and automated test harnesses).
- Integration with real banking networks or live SMS gateways.

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2. Service Decomposition

2.1 Service Responsibility & Ownership Table

Service Name	Port	Database	Primary Business Responsibility	Owned Entities
Order Service	`8001`	`orders.db`	Coordinates the checkout saga, manages customer orders, and persists order state lifecycle transitions.	`Order`, `OrderStateLog`, `IdempotencyRecord`
Restaurant Service	`8002`	`restaurant.db`	Manages restaurant catalogs, menu items, dish pricing, and handles atomic stock reservations.	`Restaurant`, `MenuItem`, `Reservation`
Payment Service	`8003`	`payment.db`	Processes payments, provides cryptographic idempotency guarantees, handles refunds, and simulates gateway failures.	`PaymentTransaction`, `IdempotencyRecord`
Delivery Service	`8004`	`delivery.db`	Tracks delivery riders, manages driver availability, dispatches tasks, and handles logistics cancellations.	`DeliveryRider`, `DeliveryTask`
Notification Service	`8005`	`notification.db`	Dispatches customer and merchant	`NotificationLog`

Service Name	Port	Database	Primary Business Responsibility	Owned Entities
			alerts (SMS, Email, Push) with failure tolerance.	

2.2 Boundary Rationale & Rejected Alternatives

- **Why this decomposition**: Each service aligns with an independent business capability (Domain-Driven Design bounded context). The rate of change and scalability characteristics of inventory (high read/write contention during flash deals) are decoupled from payment processing and physical logistics dispatch.
- **Rejected Alternative 1: Monolithic Order + Inventory Service**:
  - *Why rejected*: Coupling order placement with restaurant inventory creates a single bottleneck. Restaurants manage menus independently of orders. In Assignment 2, Restaurant Service may scale to 5 replicas during lunch peaks while Payment Service maintains fixed connections to upstream payment gateways.
- **Rejected Alternative 2: Database Layer / Controller Services**:
  - *Why rejected*: Splitting services by technical tier (e.g. “Database Service”, “Validation Service”) violates microservice cohesion principles and results in an anti-pattern distributed monolith.

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3. Architecture & Transaction Workflows

3.1 Component Architecture Diagram

```
flowchart TD
    Client["Client / Automated Test Script"] -->|POST /api/v1/orders/checkout|
OrderSvc["Order Service (Port 8001)\n[Saga Orchestrator]"]

    subgraph Core Services
        OrderSvc -->|1. POST /reservations| RestSvc["Restaurant Service (Port 8002)"]
        OrderSvc -->|2. POST /charge| PaySvc["Payment Service (Port 8003)"]
        OrderSvc -->|3. POST /tasks| DelivSvc["Delivery Service (Port 8004)"]
        OrderSvc -->|4. POST /send| NotifSvc["Notification Service (Port 8005)"]
    end

    subgraph Isolated Storage (Database-Per-Service)
        OrderSvc --- DB1[("orders.db")]
        RestSvc --- DB2[("restaurant.db")]
        PaySvc --- DB3[("payment.db")]
        DelivSvc --- DB4[("delivery.db")]
        NotifSvc --- DB5[("notification.db")]
    end
```

3.2 Sequence Diagram: Main Business Transaction (Happy Path)

```
sequenceDiagram
    autonumber
    actor Client
    participant Order as Order Service (8001)
    participant Rest as Restaurant Service (8002)
    participant Pay as Payment Service (8003)
    participant Deliv as Delivery Service (8004)
    participant Notif as Notification Service (8005)

    Client->>Order: POST /api/v1/orders/checkout (Idempotency-Key: X)
    Note over Order: State: PENDING

    Order->>Rest: POST /api/v1/restaurants/reservations
    Note over Rest: Atomic stock decrement (BR1)
    Rest-->>Order: 201 Created (reservation_id: R1)
    Note over Order: State: STOCK_RESERVED

    Order->>Pay: POST /api/v1/payments/charge (Idempotency-Key: pay-01)
    Note over Pay: Process charge & cache response (BR2)
    Pay-->>Order: 201 Created (payment_id: P1)
    Note over Order: State: PAID

    Order->>Deliv: POST /api/v1/deliveries/tasks
    Note over Deliv: Assign available rider -> BUSY (BR5)
    Deliv-->>Order: 201 Created (task_id: T1)
    Note over Order: State: DELIVERY_ASSIGNED

    Order->>Rest: POST /api/v1/restaurants/reservations/R1/confirm
    Rest-->>Order: 200 OK (Confirmed)

    Order->>Notif: POST /api/v1/notifications/send
    Notif-->>Order: 201 Created (Dispatched SMS)

    Note over Order: State: CONFIRMED
    Order-->>Client: 201 Created (Order confirmed)
```

3.3 Sequence Diagram: Partial Failure & SAGA Compensation (Failure Path)

```
sequenceDiagram
    autonumber
    actor Client
    participant Order as Order Service (8001)
    participant Rest as Restaurant Service (8002)
    participant Pay as Payment Service (8003)

    Client->>Order: POST /api/v1/orders/checkout (card ending in 9999)
    Note over Order: State: PENDING

    Order->>Rest: POST /api/v1/restaurants/reservations
    Rest-->>Order: 201 Created (Stock reserved, qty: 3)
    Note over Order: State: STOCK_RESERVED

    Order->>Pay: POST /api/v1/payments/charge
    Note over Pay: Card declined (BR4 / Condition e)
    Pay-->>Order: 402 Payment Required ("Card declined")

    Note over Order: SAGA COMPENSATION TRIGGERED<br/>State: COMPENSATING_ROLLBACK

    Order->>Rest: POST /api/v1/restaurants/reservations/R1/release
    Note over Rest: Atomic stock restoration (+3 qty)
    Rest-->>Order: 200 OK (Status: RELEASED)

    Note over Order: State: FAILED_PAYMENT_REJECTED
    Order-->>Client: 402 Payment Required (Consistent state restored)
```

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4. Data Design & Isolation

4.1 Cross-Service Reference Strategy

To strictly maintain **Database-per-Service** separation: * No shared tables or views exist across databases. * Services store only **Foreign Identifiers** (UUID strings) of entities owned by other services (e.g., `Order.customer_id`, `Order.restaurant_id`, `Order.item_id`, `Order.payment_id`). * Cross-service entity verification occurs via REST APIs at transaction time, never via database joins.

4.2 Entity Relationship Diagrams (Per Service)

```
erDiagram
    %% ORDER SERVICE
    ORDER {
        string id PK
        string customer_id
        string restaurant_id
        string item_id
        int quantity
        float unit_price
        float total_amount
        string status
        string delivery_address
        string reservation_id
        string payment_id
        string delivery_task_id
        datetime created_at
    }
    ORDER_STATE_LOG {
        string id PK
        string order_id FK
        string from_state
        string to_state
        string note
        datetime created_at
    }
    ORDER ||--o{ ORDER_STATE_LOG : tracks

    %% RESTAURANT SERVICE
    RESTAURANT {
        string id PK
        string name
        string address
        boolean is_active
    }
    MENU_ITEM {
        string id PK
        string restaurant_id FK
        string name
        float price
        int available_quantity
        int version
    }
    RESERVATION {
        string id PK
        string order_id
        string item_id FK
        int quantity
        string status
    }
    RESTAURANT ||--o{ MENU_ITEM : contains
    MENU_ITEM ||--o{ RESERVATION : reserves
```

```
%% PAYMENT SERVICE
PAYMENT_TRANSACTION {
    string id PK
    string order_id
    string customer_id
    float amount
    string status
    string payment_method
}
PAYMENT_IDEMPOTENCY {
    string id PK
    string key UK
    string request_hash
    string status
    int response_code
    text response_body
}

%% DELIVERY SERVICE
DELIVERY_RIDER {
    string id PK
    string name
    string phone
    string status
}
DELIVERY_TASK {
    string id PK
    string order_id
    string rider_id FK
    string status
}
DELIVERY_RIDER ||--o{ DELIVERY_TASK : performs
```

5. API Design & OpenAPI Contracts

5.1 Endpoint Inventory

Service	Method	Path	Status Codes	Descript
All Services	`GET`	`/health`	200	Liveness readines
Restaurant	`GET`	`/api/v1/restaurants`	200	List activ restaurar
Restaurant	`POST`	`/api/v1/restaurants`	201, 422	Register
Restaurant	`GET`	`/api/v1/restaurants/{id}/menu`	200, 404	List menu available
Restaurant	`POST`	`/api/v1/restaurants/{id}/menu`	201, 404	Add dish
Restaurant	`POST`	`/api/v1/restaurants/reservations`	201, 409, 404	Atomic s reservati (Challeng
Restaurant	`POST`	`/api/v1/restaurants/reservations/{id}/release`	200, 404	Saga compens release (3C)
Payment	`POST`	`/api/v1/payments/charge`	201, 402, 409	Idempote (Challeng
Payment	`POST`	`/api/v1/payments/{id}/refund`	200, 404	Saga compens refund
Delivery	`GET`	`/api/v1/deliveries/riders`	200	List avail riders
Delivery	`POST`	`/api/v1/deliveries/riders`	201	Register
Delivery	`POST`	`/api/v1/deliveries/tasks`	201, 409	Dispatch to availab
Delivery	`POST`	`/api/v1/deliveries/tasks/{id}/cancel`	200, 404	Compens rider rele
Order	`POST`	`/api/v1/orders/checkout`	201, 402, 409	Multi-ser Checkou

Service	Method	Path	Status Codes	Descript
Order	`GET`	`/api/v1/orders/{id}`	200, 404	Order qu state log
Notification	`POST`	`/api/v1/notifications/send`	201, 503	Send custome alert

5.2 Uniform JSON Error Model

Every service implements identical error responses:

```
{
  "error": {
    "code": "STOCK_INSUFFICIENT",
    "message": "Stock insufficient: requested 1, only 0 remaining.",
    "details": {}
  },
  "correlation_id": "corr-conc-06-a831ef"
}
```

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6. Engineering Challenges: Deep-Dive Analysis

Challenge 1: Concurrency Control (Mandatory)

- **Problem**: Preventing inventory over-allocation when 20 simultaneous threads request a dish with capacity 5.
- **Design Choice**: Atomic Conditional Database Update (`UPDATE menu_items SET available_quantity = available_quantity - :qty WHERE available_quantity >= :qty`).
- **Why Application Check-Then-Update Fails**: Check-then-update exhibits a classic read-modify-write hazard. Multiple concurrent worker processes read identical uncommitted stock, evaluate the quantity as valid, and write deductions concurrently, yielding severe negative inventory.
- **Evidence**:
  - Script: `scripts/test_challenge1_concurrency.py`
  - Execution: Fired 20 concurrent threads against 5 stock units. Exactly 5 returned HTTP 201, exactly 15 returned HTTP 409. Final stock verified at 0. (See `evidence/challenge-1-concurrency/concurrency_run_output.txt`).

Challenge 2: Idempotency Enforcement (Mandatory)

- **Problem**: Preventing double charges or duplicate orders when clients retry unsafe operations over unreliable networks.
- **Design Choice**: Header `Idempotency-Key` + SHA-256 Request Fingerprinting + Unique DB Constraint.

- **Why Sequential-Only Testing is Insufficient**: Two duplicate network packets can arrive at the server concurrently. A plain `SELECT key` before `INSERT` suffers from a race condition. By placing a database unique constraint on `idempotency_records.key`, the database enforces serialization.
- **Evidence**:
  - Script: `scripts/test_challenge2_idempotency.py`
  - Execution: Tested sequential repeat, payload alteration rejection (HTTP 409), and 10 parallel simultaneous requests. Database audit proved exactly 1 charge record created. (See `evidence/challenge-2-idempotency/idempotency_run_output.txt`).

Challenge 3C: Orchestrated Saga with Compensation (Elective)

- **Problem**: Maintaining distributed consistency across isolated microservice databases without distributed 2-Phase Commit locks.
- **Design Choice**: Centralized Orchestrator in Order Service with persistent state machine logs and automated compensating reversal actions.
- **Evidence**:
  - Script: `scripts/test_challenge3_saga.py`
  - Scenarios Tested:
    1. Happy Path: `PENDING` → `STOCK_RESERVED` → `PAID` → `DELIVERY_ASSIGNED` → `CONFIRMED`.
    2. Step 2 Payment Failure: Card declined → `COMPENSATING_ROLLBACK` releases reserved restaurant stock → `FAILED_PAYMENT_REJECTED`. Stock restored from 5 back to 8.
    3. Step 3 Delivery Failure: No riders → `COMPENSATING_ROLLBACK` refunds payment and releases stock → `FAILED_NO_RIDER_AVAILABLE`. Both payments and stock restored. (See `evidence/challenge-3-saga/saga_run_output.txt`).

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7. Testing Strategy & Results

1. **Unit Testing**: 20 unit tests across all 5 services covering business rules BR1 to BR7 (`pytest services/`).
   - **Result**: **20 passed in 0.38s**.
2. **End-to-End Testing**: `scripts/test_e2e_flow.py` verifies full lifecycle across running services.
   - **Result**: **100% success**.

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8. Architectural Decisions & Trade-Offs

Decision 1: Database-per-Service vs. Shared Database

- **Chosen**: Database-per-Service (SQLite / PostgreSQL per service).
- **Rationale**: Eliminates hidden coupling, ensures schema changes in one service do not break others, and aligns with Docker/Kubernetes pod isolation in Assignment 2.
- **Trade-off**: Requires Saga rollbacks rather than native database transactions.

Decision 2: Orchestrated Saga vs. Choreographed Saga

- **Chosen**: Orchestrated Saga (Order Service as coordinator).
- **Rationale**: Order Service retains explicit, centralized visibility over the transaction state machine. Compensating actions and failure diagnostics are straightforward to debug and trace.
- **Trade-off**: Order Service has coupling to the APIs of downstream services.

Decision 3: Atomic Conditional SQL Updates vs. Distributed Redis Locks

- **Chosen**: Atomic Conditional SQL Update (`available_quantity >= :qty`).
- **Rationale**: Leverages the ACID guarantees of the relational engine without introducing external distributed locking dependencies, lock TTL expiration bugs, or split-brain hazards.
- **Trade-off**: Database handles the contention serialization.

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9. Limitations & Future Work

- **Limitations**: Current inter-service communication is synchronous REST. High network latencies compound across multi-step transactions.
- **Future Work (Assignment 2)**: Containerization with Docker Compose and Minikube; integration of message broker (RabbitMQ/Kafka) for asynchronous notification dispatch.

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10. Links & Artifacts

- **GitHub Repository**: https://github.com/kush6542/Group26_Scalable_Services
- **Google Drive (Demo Video & Report PDF)**: [Drive Folder Link]

Appendix A: Video Presentation Outline (8 to 10 Minutes)

- **0:00 - 1:00 (1 min)**: Team introduction, service ownership, domain overview.
- **1:00 - 3:00 (2 min)**: Architectural walkthrough: 5 services, Database-per-Service, state machine.
- **3:00 - 6:00 (3 min)**: Live demo of End-to-End checkout (happy path) and simulated card decline (failure compensation).
- **6:00 - 9:00 (3 min)**: Live execution of Challenge scripts:
  - Challenge 1: 20 parallel threads vs capacity 5.
  - Challenge 2: Idempotent replays and payload conflict.
  - Challenge 3C: Saga compensating multi-service rollbacks.
- **9:00 - 10:00 (1 min)**: Wrap-up, trade-offs, and transition to Assignment 2 containerization.

Appendix B: Student Contributions & Reflections

B.1 Contribution Matrix

Student Name	Student ID / Email	Service Owned	Key Contributions
Kushagra Parashar	`2025tm93126@wilp.bits-pilani.ac.in`	**Order Service** & **Restaurant Service**	Architected the core multi-service checkout Saga Orchestrator, persistent state machine, and Challenge 1 atomic SQL concurrency control (`available_quantity >= :qty`). Built end-to-end integration test harness and unified service runner.
Thamarai Muthu Selvi A	`2025tm93028@wilp.bits-pilani.ac.in`	**Payment Service**	Implemented Challenge 2 idempotency engine with cryptographic SHA-256 request payload hashing, database unique constraints for concurrent race protection, and simulated failure injection for Saga rollbacks.
Vijay Kumar Singh	`2025tm93550@wilp.bits-pilani.ac.in`	**Delivery Service**	Engineered the rider registry, rider status locking (`AVAILABLE` -> `BUSY`), delivery task allocation lifecycle, and compensating cancellation endpoints for multi-step Saga rollbacks.
Hemanta Kumar Rana	`2025tm93157@wilp.bits-pilani.ac.in`	**Notification Service**	Developed the multi-channel notification dispatching service (SMS, Email, Push), third-party gateway failure simulation, health probe

Student Name	Student ID / Email	Service Owned	Key Contributions
			monitoring, and distributed correlation ID tracing middleware.

B.2 Individual Reflections

1. Kushagra Parashar (Order Service & Restaurant Service)

- **What I Built**: I developed the Order Service and the Restaurant Service. In Order Service, I built the centralized Saga Orchestrator that coordinates the distributed business transaction across four independent services and logs explicit state machine transitions (`PENDING`, `STOCK_RESERVED`, `PAID`, `DELIVERY_ASSIGNED`, `CONFIRMED`, `COMPENSATING_ROLLBACK`). In Restaurant Service, I implemented the catalog, menu inventory, and atomic conditional stock updates for Challenge 1.
- **Hardest Problem Solved**: Eliminating race conditions under simultaneous bursts. When 20 threads hit a single menu item with capacity 5, naive application code ("check-then-update") caused extreme over-allocation. I solved this by pushing the concurrency check into an atomic SQL `UPDATE ... WHERE available_quantity >= :qty` statement, mathematically preventing overselling while avoiding coarse table locks.
- **What I Would Change**: In future iterations (Assignment 2), I would introduce an asynchronous message broker (such as Kafka or RabbitMQ) to decouple the Saga steps, transitioning from synchronous HTTP orchestration to event-driven choreography for non-blocking scale.

2. Thamarai Muthu Selvi A (Payment Service)

- **What I Built**: I designed and developed the Payment Service, including charge processing, compensating refund endpoints, and the complete Idempotency Engine (Challenge 2) to prevent double debits during network retries or duplicate clicks.
- **Hardest Problem Solved**: Handling concurrent duplicate hits arriving at the exact same millisecond. A standard check-then-store lookup fails under simultaneous identical requests due to read-commit latencies. I solved this by adding a database-level unique constraint on `idempotency_records.key` paired with `IN_PROGRESS` state tracking, guaranteeing that exactly one transaction executes while concurrent duplicates await the cached result.
- **What I Would Change**: I would add time-to-live (TTL) expiration on the idempotency cache records and implement automated token bucket rate-limiting to protect payment gateways from distributed denial-of-service traffic.

3. Vijay Kumar Singh (Delivery Service)

- **What I Built**: I implemented the Delivery Service responsible for tracking rider states (`AVAILABLE`, `BUSY`, `OFFLINE`), dispatching delivery tasks to eligible couriers, and providing a compensating `POST /tasks/{id}/cancel` endpoint.
- **Hardest Problem Solved**: Managing the logistics compensation lifecycle during Saga rollbacks. When downstream operations fail, the delivery task must not only be marked

`CANCELLED`, but the assigned rider must be atomically restored to `AVAILABLE` status without creating orphaned driver states.
- **What I Would Change**: I would incorporate geo-spatial proximity indexing (e.g., PostGIS) so that riders are dispatched based on real-time Euclidean or road distance to the pickup restaurant rather than first-available FIFO order.

4. Hemanta Kumar Rana (Notification Service)

- **What I Built**: I developed the Notification Service handling SMS, email, and push alerts with persistent dispatch logging, simulated gateway unreachability (HTTP 503), and distributed tracing middleware.
- **Hardest Problem Solved**: Establishing end-to-end observability and log correlation. In a distributed multi-service setup, debugging errors without unified identifiers is impossible. I helped implement the `CorrelationIdMiddleware` that extracts or generates `X-Correlation-ID` and propagates it across all service-to-service HTTP headers and log streams.
- **What I Would Change**: I would implement an outbox pattern with dead-letter queue (DLQ) retry mechanisms so that temporary notification provider failures do not lose pending alerts.

Appendix C: AI-Use Declaration

AI assistance (Google Gemini / Antigravity) was utilized for generating boilerplate code scaffolding, writing repetitive test client assertions, and formatting documentation. All core architectural decisions, concurrency mechanics, idempotency hashing algorithms, and distributed Saga state machines were reviewed, verified, and executed by the team.